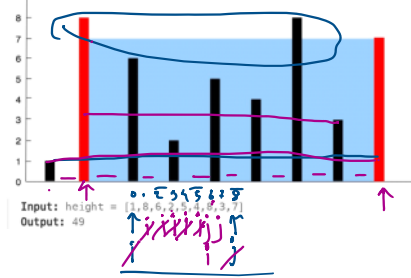
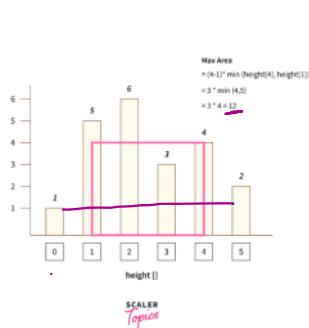


Example 1:



① 7

$$\begin{aligned}
 j-i \\
 8-0 &= 8 * 1 = 8 \\
 8-1 &= 7 * 7 = 49 \\
 7-1 &= 6 * 3 = 18 \\
 6-1 &= 5 * 2 = 10 \\
 6-2 &= 4 * 6 = 24 \\
 6-3 &= 3 * 2 = 6 \\
 6-4 &= 2 * 5 = 10 \\
 6-5 &= 1 * 4 = 4
 \end{aligned}$$



[⁰1, ¹5, ²6, ³3, ⁴4, ⁵2]

$$5-0 = 5 * 1 = 5$$

[4, 3, 2, 9] → 4

$$\begin{array}{r}
 9 \\
 \times 1 \\
 \hline
 9
 \end{array}$$

[1, 0]

$$\begin{array}{r}
 4321 \\
 \times 1 \\
 \hline
 4321
 \end{array}$$

[4, 3, 2, 2] → 4

[4, 3, 2, 1]

$$4 * 10 \% 10 = 4$$

[2, 2, 3, 4]

$$\begin{array}{r}
 4 * 10 \rightarrow 40 \\
 \quad 3 \\
 \hline
 43 \\
 * 10 \\
 \hline
 430 \\
 \quad 2 \\
 \hline
 432 \\
 * 10 \\
 \hline
 4320 \\
 \quad 1 \\
 \hline
 4321
 \end{array}$$

4, 3, 2, 2

4320
4321
4322

$[9, 9] \rightarrow 99$
 $+ 1$
 $\hline 100$
 $\rightarrow [1, 0, 0]$
~~1 0 0~~
 $\rightarrow [0, 0, 0] \rightarrow \text{count} = 3$
 $\uparrow \quad \uparrow \quad \uparrow$

9876543210
9876543211

```
int value = arr[0];
for(int i = 1 ; i < arr.length ; i++){
    value *= 10;
    value += arr[i];
}
value++;
int copyVal = value;
System.out.println(value);
int count = 0;
while(copyVal != 0){
    count++;
    copyVal /= 10;
}
int newArr[] = new int[count];
count--;
while(value != 0){
    int digit = value%10;
    newArr[count--] = digit;
    value /= 10;
}
```

[0]
 $\uparrow \quad \uparrow$

[4, 3, 3, 0] [4, 3, 3, 0]
 $\uparrow \quad \uparrow \quad \uparrow$

[1, 0]
 $\uparrow$

[0, 0, 0] $\rightarrow$ [1, 0, 0, 0]
 $\uparrow \quad \uparrow \quad \uparrow \quad \uparrow$

↑
[8, 8, 0]
↑ ↑

[8, 8, 0]

↑ ↑ ↑

[0, 0, 0, 0]³⁺¹

↑
[1, 0, 0, 0]

[1, 0, 9, ~~8~~, ~~0~~]

→ [1, 0, 9, 9, 0]

✗
for(int i = arr.length-1; i>=0; i--){
if(arr[i] != 9){
arr[i]++;
return arr;
}
arr[i] = 0;
}

✓
int newArr [] = new int[arr.length+1];
newArr[0] = 1;
return newArr;

↑ [0]
↑ [0, 0]
↑

[1, 0]

[1, 2, 3]
[1, 3, 2]
→ [2, 1, 3]
[2, 3, 1]
[3, 1, 2]
[3, 2, 1]

[1, 2, 3]
[3, 2, 1]
[2, 3, 1]
[2, 1, 3]